Executive Briefing & Facility Vulnerability Report

CRITICAL SECURITY AUDIT FINDING: 100% UNREGISTERED / UNKNOWN SUBJECTS
Zero facial profiles match an enrolled employee database. **All 1,652 detected personas (SUBJ-01 to SUBJ-1652) represent UNKNOWN PERSONS (Pending Identification).** Tiers reflect sighting frequency only—NOT confirmed employment or authorized clearance.

1. Executive Mission Overview

An autonomous forensic and biometric surveillance audit was conducted over 30 continuous calendar days of native 3K CCTV footage covering the RO Water Treatment Plant rooftop. The audit established continuous tracking of all human activity in the mechanical wing, measured shift-by-shift recurrence patterns, resolved historical video clipping blindspots, and logged every direct physical contact event against mission-critical equipment assets.

2. Key Operational Metrics

Metric Category	Audit Result	Operational Security Context
Total Facility Coverage	**713.66 Recorded Hours**	30 continuous calendar days (100% of Sept 2026 footage)
Total Discovered Personas	**1,652 Unique Personas**	100% Unknown / Unregistered Individuals grouped via ArcFace clustering
High-Frequency Unknowns	**12 Individuals**	Tier 1 recurring unknown subjects (50+ detections each)
Direct Equipment Contacts	**18 Unknown Individuals**	Verified touching multiport valves, skid controls, or electrical breaker switch
Pristine Incident Clips	**179 Events (8.20 GB)**	Lossless H.265 stream cuts with +/-60s pre/post buffers
Processed Face Vectors	**36,226 Embeddings**	512-D ArcFace vectors passing strict quality threshold (Q >= 40)

3. Operational Shift Workload Distribution

Shift A (Morning, 06:00 – 13:59): 24,625 detections (68.0%) — Peak operating window where recurring unknown subjects routinely operated valves.
Shift B (Afternoon, 14:00 – 21:59): 11,188 detections (30.9%) — Secondary window with multiple unknown subjects traversing the corridor.
Shift C (Night Vigilance, 22:00 – 05:59): 413 detections (1.1%) — Night visits, perimeter walkthroughs, and main breaker switch contacts.

4. 30-Day Critical & Severe Incident Investigations (Days 01 – 30)

Across the 30 continuous calendar days (713.66 recorded hours), **179 physical equipment interaction events** were captured and investigated across all four operational weeks:

Day & Time	Event ID	Facility Asset	Unknown Subjects	Duration
Day 09 16:21:55	EVT-D09-005	Power Breaker Switch (Zone 2)	**SUBJ-233, SUBJ-759, SUBJ-760**	162s
Day 09 16:24:51	EVT-D09-006	Power Breaker Switch (Zone 2)	**SUBJ-13, SUBJ-347**	125s
Day 11 06:33:37	EVT-D11-001	Power Breaker Switch (Zone 2)	**SUBJ-791**	121s
Day 24 16:39:26	EVT-D24-013	Power Breaker Switch (Zone 2)	**SUBJ-1079**	121s
Day 25 11:43:10	EVT-D25-002	Power Breaker Switch (Zone 2)	**SUBJ-1088**	121s

Critical Multiport Backwash Valve Operations (Zone 1A/B — 54 Events across 23 Days):
• **Week 1 (Days 01–07, 9 events):** Routine morning backwashes, including Day 1 events (SUBJ-01 at 07:29/08:14 and SUBJ-1652 at 09:53:38).
• **Week 2 (Days 08–14, 13 events):** Daily operational cycling by Tier 1 and Tier 2 unknown subjects.
• **Week 3 (Days 15–21, 17 events):** Peak critical activity window, featuring Day 15 multi-subject group access (5 valve operations involving 10 unknown subjects).
• **Week 4 (Days 22–26, 15 events):** Frequent high-contact days including Day 25 (4 valve operations) and Day 26 mass gathering (3 events, 34 unknown subjects on deck).
• **Night Vigilance (Shift C):** Off-hours visits on Day 01 (22:27:36), Day 15 (23:24:04), and Day 29 (23:14:10).

5. Forensic Pipeline Bug Resolutions & Universal 30-Day Fixes

Day 1 served as the forensic calibration testbed, uncovering software failure modes that prompted universal fixes across all 30 days:
1. Premature Motion Cutoff: Clip 14 automated scanner timed out after 42s, missing the 09:53:38 valve manipulation. Resolved via rolling temporal windowing and cataloging the unknown individual as **SUBJ-1652** (verified non-match to SUBJ-02, cosine sim 0.1831).
2. File Split Boundary Truncation: Split boundary at 22:28:00 cut off the Day 1 22:27:36 breaker contact. Resolved by enforcing mandatory **±60s contextual safety buffers** bridging files across all 179 clips.
3. SQL Inversion Bug: Fixed SQL timestamp ordering across all 30 clipping indices (0 inversions across all 179 events).

6. Immediate Security Recommendations

1. Mandatory Personnel Enrollment: Conduct immediate formal identity verification with facility HR to associate names and employment status with all 1,652 unknown subject clusters.
2. Physical Access Restriction: Secure Zone 1A/B Valves and Zone 2 Power Switch with physical lockout/tagout (LOTO) padlock boxes.
3. Real-Time Spatial Alerting: Implement automated audible alerts when any unauthorized centroid breaches the 40px buffer of Zone 1 or Zone 2.